WORKED SOLUTION

STEP I 2011, Question 5

Pure • Integrals with Modulus

STEP I 2011, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q5

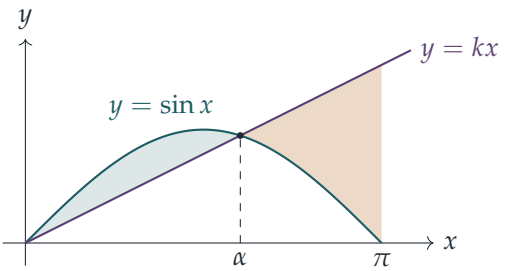
The key idea

The modulus sign is only awkward until we know where $\sin x - kx$ changes sign, and the sketch tells us: once, at $x = \alpha$. Split the integral there and the modulus disappears. The last step is to stop thinking of k as the variable: since $k = \sin\alpha/\alpha$, everything can be written in terms of α, and the derivative of I then factorises neatly.

Solution

The unique root

Let $g(x) = \sin x - kx$. The line $y = kx$ passes through the origin with gradient $k < 1$, while $y = \sin x$ leaves the origin with gradient $\cos 0 = 1$. So just to the right of 0 the sine curve is above the line. At $x = \pi$ the sine curve is back at 0 but the line is at height $k\pi > 0$, so there the line is above. Since $y = \sin x$ bends downwards on $(0, \pi)$ and the line is straight, they cross exactly once in between.



To make "exactly once" watertight: $g(0) = 0$, $g'(x) = \cos x - k$ and $g''(x) = -\sin x < 0$ on $(0, \pi)$. So g' is strictly decreasing, from $1 - k > 0$ at 0 to $-1 - k < 0$ at π, and vanishes at exactly one point β. Hence g is strictly increasing on $(0, \beta]$, so positive there (it starts at 0), and strictly decreasing on $[\beta, \pi)$, from a positive value to $g(\pi) = -k\pi < 0$.

It therefore has exactly one zero α in $(0, \pi)$, and

$$\sin x > kx \text{ for } 0 < x < \alpha, \qquad \sin x < kx \text{ for } \alpha < x < \pi.$$

Evaluating I

Using these signs (the teal and amber regions in the sketch),

$$I = \int_0^{\alpha} (\sin x - kx)\,\mathrm{d}x - \int_{\alpha}^{\pi} (\sin x - kx)\,\mathrm{d}x.$$

An antiderivative of $\sin x - kx$ is $F(x) = -\cos x - \frac{1}{2}kx^2$, so

$$\begin{aligned} I &= (F(\alpha) - F(0)) - (F(\pi) - F(\alpha)) = 2F(\alpha) - F(0) - F(\pi) \\ &= (-2\cos\alpha - k\alpha^2) + 1 - (1 - \tfrac{1}{2}k\pi^2) \\ &= \tfrac{1}{2}k\pi^2 - 2\cos\alpha - k\alpha^2. \end{aligned}$$

Since α solves $(*)$, $k = \dfrac{\sin\alpha}{\alpha}$, so $k\alpha^2 = \alpha\sin\alpha$ and

$$I = \frac{\pi^2 \sin\alpha}{2\alpha} - 2\cos\alpha - \alpha\sin\alpha,$$

as required.

The stationary value

? **Ask yourself** What values can α take?

Each $k \in (0, 1)$ gives an $\alpha \in (0, \pi)$. Conversely, each $\alpha \in (0, \pi)$ comes from $k = \sin\alpha/\alpha$, which lies in $(0, 1)$ because $0 < \sin\alpha < \alpha$ there. So we must study $I(\alpha)$ for all α in the open interval $(0, \pi)$.

Differentiating, with the quotient rule on the first term,

$$\begin{aligned} \frac{\mathrm{d}I}{\mathrm{d}\alpha} &= \frac{\pi^2(\alpha\cos\alpha - \sin\alpha)}{2\alpha^2} + 2\sin\alpha - (\sin\alpha + \alpha\cos\alpha) \\ &= \frac{\pi^2(\alpha\cos\alpha - \sin\alpha)}{2\alpha^2} - (\alpha\cos\alpha - \sin\alpha) \\ &= (\alpha\cos\alpha - \sin\alpha)\left(\frac{\pi^2}{2\alpha^2} - 1\right). \end{aligned}$$

The first factor is never zero on $(0, \pi)$. Let $h(\alpha) = \alpha\cos\alpha - \sin\alpha$. Then $h(0) = 0$ and $h'(\alpha) = \cos\alpha - \alpha\sin\alpha - \cos\alpha = -\alpha\sin\alpha < 0$ on $(0, \pi)$, so $h(\alpha) < 0$ throughout. (Equivalently, $\tan\alpha > \alpha$ on $(0, \frac{1}{2}\pi)$, and $\cos\alpha \le 0$ beyond.)

So $\mathrm{d}I/\mathrm{d}\alpha = 0$ exactly when $\pi^2 = 2\alpha^2$, that is, at $\alpha = \pi/\sqrt{2}$, which does lie in $(0, \pi)$. This is the unique stationary value. Since $h < 0$, the sign of $\mathrm{d}I/\mathrm{d}\alpha$ is opposite to the sign of $\frac{\pi^2}{2\alpha^2} - 1$:

- for $0 < \alpha < \pi/\sqrt{2}$, $\frac{\pi^2}{2\alpha^2} > 1$, so $\mathrm{d}I/\mathrm{d}\alpha < 0$;
- for $\pi/\sqrt{2} < \alpha < \pi$, $\frac{\pi^2}{2\alpha^2} < 1$, so $\mathrm{d}I/\mathrm{d}\alpha > 0$.

So I decreases and then increases: the stationary value is a minimum, and since it is the only stationary point on the interval, it is the smallest value of I.

At $\alpha = \pi/\sqrt{2}$ we have $\dfrac{\pi^2}{2\alpha} = \dfrac{\pi^2}{\sqrt{2}\,\pi} = \dfrac{\pi}{\sqrt{2}} = \alpha$, so the first and third terms of I cancel:

$$I_{\min} = \alpha \sin\alpha - 2\cos\alpha - \alpha\sin\alpha = -2\cos\frac{\pi}{\sqrt{2}}.$$

This is positive, as it must be, since $\pi/\sqrt{2} \approx 2.22$ lies between $\frac{1}{2}\pi$ and π, where cosine is negative.

Checks

- As $k \to 0$, $\alpha \to \pi$ and the formula gives $I \to 2$, which is $\int_0^\pi \sin x\,\mathrm{d}x$. As $k \to 1$, $\alpha \to 0$ and $I \to \frac{1}{2}\pi^2 - 2$, which is $\int_0^\pi (x - \sin x)\,\mathrm{d}x$.
- Numerically, $-2\cos(\pi/\sqrt{2}) \approx 1.2114$, which is less than both end values 2 and $\frac{1}{2}\pi^2 - 2 \approx 2.935$. Direct midpoint quadrature of $\int_0^\pi |\sin x - kx|\,\mathrm{d}x$ with $k = \sin\alpha/\alpha \approx 0.3582$ also gives 1.2114.

Common slips

- Differentiating with respect to α while treating k as a constant: k depends on α, which is exactly why we substitute $k = \sin\alpha/\alpha$ first.
- Showing $\mathrm{d}I/\mathrm{d}\alpha = 0$ only at $\alpha = \pi/\sqrt{2}$ but forgetting to show that $\alpha\cos\alpha - \sin\alpha$ has no zero in $(0, \pi)$.

Beyond STEP

The argument for the unique root uses the intermediate value theorem together with strict monotonicity, and the minimisation quietly uses the fact that $\alpha \mapsto k$ is a bijection between $(0, \pi)$ and $(0, 1)$, so that minimising over α is the same as minimising over k. Making such reparametrisations and existence arguments rigorous is the business of a first course in real analysis; Abbott's *Understanding Analysis* is an excellent place to start.

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